

RAPID DECAY AND C^* -SIMPLICITY: A SUPPLEMENTARY SUNFLOWER PROOF

CALEB BARNETT

ABSTRACT. This supplementary note accompanies the paper of Barnett, Gondek, and Kalantar proving that a finitely generated group with rapid decay and trivial amenable radical is C^* -simple. The joint paper gives the primary, human-written proof of the result. Here we record an alternative sunflower proof generated with a language model and independently checked by the author. Failure of simplicity gives an amenable subgroup meeting every conjugate of a fixed finite set. We pack arbitrarily many disjoint conjugates of this set into a ball whose radius is logarithmic in their number, contradicting the growth bound on amenable subgroups imposed by rapid decay. The alternative argument also applies to sub-rapid decay. The closing disclosure documents its development, verification, and use of tools.

1. INTRODUCTION

Barnett, Gondek, and Kalantar give a human-written proof that a finitely generated group with rapid decay and trivial amenable radical is C^* -simple, answering the rapid-decay part of Question 1 of Alam, Gondek, Kalantar, and Pham [AGKP, §3, Question 1]. Their joint paper is the primary reference for this result. The present note is a supplement: it records an AI-generated alternative proof using the sunflower lemma, together with an account of how that argument was developed and checked. We restate the rapid-decay conclusion to make the supplement self-contained.

Theorem 1.1. *A finitely generated group with rapid decay and trivial amenable radical is C^* -simple.*

The starting observation is the following. If an element lies outside the amenable radical, its normal closure is nonamenable. Since amenability is preserved under directed unions, finitely many conjugates of that element already generate a nonamenable subgroup. Applying the tuple argument from [AGKP, proof of Proposition 10(ii)] to this subgroup shows that there are exponentially many distinct conjugates obtainable by short conjugators.

The next step is to move a whole finite set by one conjugator. This is where the sunflower lemma enters. A large family of images contains many disjoint petals around a common core. We move the smaller core away from a prescribed finite set by induction; one of the petals must then avoid that set as well. Keeping track of the conjugators gives a logarithmic bound on the length of this displacement.

To see why this proves the theorem, suppose that simplicity fails. Kennedy's characterization by amenable uniformly recurrent subgroups supplies an amenable subgroup H and a finite set $P \subseteq \Gamma \setminus \{e\}$ such that

$$H \cap \gamma P \gamma^{-1} \neq \emptyset \quad (\gamma \in \Gamma).$$

The displacement argument packs N disjoint conjugates of P into a ball of radius $O_P(\log N)$. Each contributes a different element to H . Rapid decay, however, bounds the number of elements of an amenable subgroup in such a ball by a fixed power of $\log N$.

The proof separates the hypotheses quite cleanly. Triviality of the amenable radical supplies growth of the rooted conjugacy orbits. The finite-set argument is combinatorial. Rapid decay is used only for the upper bound on the growth of amenable subgroups. The last comparison also works under sub-rapid decay, as recorded in Corollary 5.1.

I independently checked the AI-generated proof presented here. The Barnett–Gondek–Kalantar paper was developed without LLM assistance after a discussion of the starting observation above. Gondek and Kalantar did not read or verify this alternative proof. The closing disclosure explains the relationship between the joint paper and this supplement, and records the development and verification of the AI-generated argument.

Notation. Throughout, Γ is a finitely generated discrete group, S is a fixed finite symmetric generating set, and $|\cdot|$ is its word length. Write

$$B_r = \{\gamma \in \Gamma : |\gamma| \leq r\}.$$

Radii are nonnegative integers, and logarithms are natural. The left regular representation on $\ell^2(\Gamma)$ is denoted by λ , and

$$\lambda(f) = \sum_{g \in \Gamma} f(g) \lambda_g \quad (f \in \mathbb{C}\Gamma).$$

The reduced group C^* -algebra $C_r^*(\Gamma)$ is the operator-norm closure of these finite sums. The group is C^* -simple if $C_r^*(\Gamma)$ has no nonzero proper closed two-sided ideals. Its *amenable radical*, denoted by $\text{Rad}(\Gamma)$, is its largest amenable normal subgroup.

Definition 1.2. The *reduced norm growth function* is

$$\rho(r) = \sup_{\substack{0 \neq f \in \mathbb{C}\Gamma \\ \text{supp } f \subseteq B_r}} \frac{\|\lambda(f)\|}{\|f\|_2}.$$

Here the numerator is the operator norm on $\ell^2(\Gamma)$ and the denominator is the coefficient norm for counting measure. The group has *rapid decay* if there are constants $C > 0$ and $d \geq 0$ such that

$$\rho(r) \leq C(1+r)^d \quad (r \geq 0). \quad (1.1)$$

It has *sub-rapid decay* if $\log \rho(r)/r \rightarrow 0$ as $r \rightarrow \infty$.

These properties do not depend on the chosen finite generating set: changing that set changes word length by at most a fixed multiplicative factor. Sub-rapid decay always refers here to the full limit.

2. THE OBSTRUCTION TO SIMPLICITY

Identify each subgroup of Γ with its characteristic function. The resulting topology makes $\text{Sub}(\Gamma)$ a compact subspace of $\{0, 1\}^\Gamma$, on which Γ acts by conjugation.

Definition 2.1. A *uniformly recurrent subgroup* (URS) is a nonempty closed conjugation-invariant subset of $\text{Sub}(\Gamma)$ which is minimal with these properties. It is *amenable* if all its members are amenable, and *nontrivial* if it is not $\{\{e\}\}$.

Kennedy’s criterion says that Γ is C^* -simple if and only if it has no nontrivial amenable URS [Ken, Theorem 4.1]. We need the following finite consequence.

Lemma 2.2. *If Γ is not C^* -simple, there are an amenable subgroup $H \leq \Gamma$ and a finite nonempty set $P \subseteq \Gamma \setminus \{e\}$ such that*

$$H \cap \gamma P \gamma^{-1} \neq \emptyset \quad (\gamma \in \Gamma). \quad (2.1)$$

Proof. Let $\mathcal{U}$ be a nontrivial amenable URS. It does not contain $\{e\}$, since minimality would then give $\mathcal{U} = \{\{e\}\}$. The clopen sets

$$\{K \in \mathcal{U} : g \in K\}, \quad g \neq e,$$

therefore cover $\mathcal{U}$. A finite subcover gives a finite nonempty set P meeting every member of $\mathcal{U}$. Fix $H \in \mathcal{U}$. For every $\gamma \in \Gamma$, the subgroup $\gamma^{-1}H\gamma$ belongs to $\mathcal{U}$, so it meets P . Conjugating this intersection by γ gives (2.1). $\square$

Minimality is used only to exclude the trivial subgroup. Compactness gives P , and conjugation invariance makes the same P work for every conjugate of H .

Lemma 2.3. *For every amenable subgroup $H \leq \Gamma$ and every $r \geq 0$,*

$$|H \cap B_r| \leq \rho(r)^2. \quad (2.2)$$

Proof. Put $E = H \cap B_r$. Let (F_j) be a Følner sequence of nonempty finite subsets of H , and regard $\xi_j = |F_j|^{-1/2}1_{F_j}$ as a unit vector in $\ell^2(\Gamma)$. Then

$$\langle \lambda(1_E)\xi_j, \xi_j \rangle = \sum_{g \in E} \frac{|gF_j \cap F_j|}{|F_j|} \longrightarrow |E|.$$

It follows that

$$|E| \leq \|\lambda(1_E)\| \leq \rho(r)\sqrt{|E|}.$$

Since $e \in E$, we can divide by $\sqrt{|E|}$ and square. $\square$

No finite generation of H is required. In particular, rapid decay gives the ambient growth bound

$$|H \cap B_r| \leq C^2(1+r)^{2d}.$$

We will contradict this bound by finding disjoint conjugates of the set in Lemma 2.2 inside a small ball.

3. SHORT CONJUGATORS ARE PLENTIFUL

For $g \in \Gamma$, write

$$\beta_g(r) = |\{\gamma g \gamma^{-1} : \gamma \in B_r\}|.$$

This is the *rooted conjugacy growth function*: it counts conjugates reached by short conjugators. It differs from the number of conjugates which happen to lie in B_r . The rooted function is useful here because

$$|\gamma g \gamma^{-1}| \leq 2|\gamma| + |g|. \quad (3.1)$$

Lemma 3.1. *If $g \notin \text{Rad}(\Gamma)$, there are constants $c_g \in (0, 1]$ and $\delta_g > 0$ such that*

$$\beta_g(r) \geq c_g e^{\delta_g r} \quad (r \geq 0).$$

Proof. The normal closure of g is nonamenable. It is the directed union of the subgroups generated by finite sets of conjugates of g . Amenability is preserved under directed unions, so some subgroup

$$A = \langle x_1, \dots, x_k \rangle, \quad x_i = u_i g u_i^{-1},$$

is nonamenable. Since $Z(A)$ is abelian and amenability is preserved under extensions, the quotient $Q = A/Z(A)$ is also nonamenable. These permanence properties of amenability can be found in [BHV, Appendix G].

Conjugating the entire generating tuple distinguishes elements of A modulo its center. More precisely, the map

$$Q \longrightarrow \Gamma^k, \quad aZ(A) \longmapsto (ax_i a^{-1})_{i=1}^k$$

is well defined and injective: $a, b \in A$ give the same tuple exactly when $b^{-1}a$ commutes with every generator of A . This is the tuple argument in [AGKP, proof of Proposition 10(ii)].

Give Q the word metric from the images of $x_1^{\pm 1}, \dots, x_k^{\pm 1}$. The Følner criterion [BHV, Corollary G.5.6] gives $q > 1$ such that multiplication by these generators and the identity expands every finite nonempty subset of Q by a factor of at least q . Iterating on balls gives

$$|B_Q(n)| \geq q^n \quad (n \geq 0).$$

Choose an integer $L \geq 1$ bounding all $|x_i|$ and $|u_i|$. A word of length at most n in the quotient generators lifts to the same word in A , giving a representative a with $|a| \leq Ln$. Each entry in its tuple has the form

$$ax_ia^{-1} = (au_i)g(au_i)^{-1}, \quad |au_i| \leq L(n+1).$$

There are at most $\beta_g(L(n+1))$ possibilities for each entry. Distinct quotient elements give distinct tuples, hence

$$q^n \leq |B_Q(n)| \leq \beta_g(L(n+1))^k. \quad (3.2)$$

The tuple length k is fixed, so taking a k th root preserves exponential growth.

For completeness, if $r \geq L$, take $n = \lfloor r/L \rfloor - 1$. Then $L(n+1) \leq r$ and $n \geq r/L - 2$, giving

$$\beta_g(r) \geq q^{n/k} \geq q^{-2/k} \exp\left(\frac{\log q}{kL}r\right).$$

For $0 \leq r < L$, the last expression is less than 1, while $\beta_g(r) \geq 1$. The same bound therefore holds at every radius. $\square$

4. MOVING A FINITE SET

For a single element outside the amenable radical, Lemma 3.1 already gives the displacement we need. To avoid an n -point set, choose a radius at which the rooted orbit has more than n points. This costs $O(\log(n+1))$ in word length. For a finite set P , all its points must be moved by the same conjugator, and moving one point can bring another back into the forbidden set.

We use the classical sunflower lemma of Erdős and Rado [ER]. A family of distinct finite sets is a *sunflower* if every pair has the same intersection, called its *core*. Removing the core leaves pairwise disjoint *petals*. The core is allowed to be empty. We include the elementary bound that will be used below.

Lemma 4.1. *Let $k, n \geq 1$. A family of more than $k!n^k$ distinct k -element sets contains a sunflower with $n+1$ members. Its core has fewer than k points.*

Proof. Induct on k . The assertion for singletons is immediate. For $k > 1$, take a maximal pairwise disjoint subfamily. If it has at least $n+1$ members, it contains an empty-core sunflower of the required size. Otherwise, its union has at most kn points and meets every set in the family. Some point therefore belongs to more than

$$\frac{k!n^k}{kn} = (k-1)!n^{k-1}$$

members. Delete that point from those members. The resulting $(k-1)$ -element sets are still distinct, so induction gives a sunflower with $n+1$ members. Restore the deleted point to obtain the desired sunflower. The core is proper because its members are distinct and have the same size. $\square$

The petals let us reduce the displacement problem to the smaller core. The length bound requires one further choice: we pull the core back into the original finite set before applying induction. We must pull the forbidden set back at the same time.

For the rest of this section, write $\gamma \cdot x = \gamma x \gamma^{-1}$ and $\gamma \cdot E = \gamma E \gamma^{-1}$.

Lemma 4.2. *Let $P \subseteq \Gamma \setminus \text{Rad}(\Gamma)$ be finite and nonempty. There is $C_P > 0$ such that, for every finite $F \subseteq \Gamma$, some $\gamma \in \Gamma$ satisfies*

$$(\gamma \cdot P) \cap F = \emptyset, \quad |\gamma| \leq C_P \log(|F| + 1).$$

Proof. If $F = \emptyset$, take $\gamma = e$. Otherwise put $m = |P|$ and $n = |F| \geq 1$. By Lemma 3.1, we can choose common constants $c \in (0, 1]$ and $\delta > 0$ such that

$$\beta_p(r) \geq ce^{\delta r} \quad (p \in P, r \geq 0).$$

Choose an integer $R \geq 1$ with

$$ce^{\delta R} > m m! n^m. \quad (4.1)$$

We can take $R = O_P(\log(n + 1))$; an explicit choice is given in Appendix A.

Keep P, n, R fixed. We prove by strong induction on k that, for every $E \subseteq P$ with $|E| = k$ and every forbidden set F of cardinality n , there is γ such that

$$(\gamma \cdot E) \cap F = \emptyset, \quad |\gamma| \leq 3kR. \quad (4.2)$$

The assertion is uniform in the positions of the forbidden points. For $k = 0$, use the identity.

Suppose $k \geq 1$ and choose $p \in E$. Let $\mathcal{A}$ be the family of distinct sets $\gamma \cdot E$ with $|\gamma| \leq R$. Each member has k points, and their union contains $B_R \cdot p$. Consequently,

$$k|\mathcal{A}| \geq \beta_p(R) > m m! n^m, \quad |\mathcal{A}| > k! n^k.$$

The last inequality uses $k \leq m$ and $n \geq 1$. Lemma 4.1 gives a sunflower

$$\gamma_1 \cdot E, \dots, \gamma_{n+1} \cdot E, \quad |\gamma_i| \leq R,$$

with core $\mathcal{C}$ of size less than k .

Since $\mathcal{C} \subseteq \gamma_1 \cdot E$, put

$$D = \gamma_1^{-1} \cdot \mathcal{C} \subsetneq E, \quad F' = \gamma_1^{-1} \cdot F.$$

We have $D \subseteq P$, $|D| < k$, and $|F'| = n$. Induction gives an element η with

$$(\eta \cdot D) \cap F' = \emptyset, \quad |\eta| \leq 3(k-1)R.$$

Set $h = \gamma_1 \eta \gamma_1^{-1}$. Then

$$h \cdot \mathcal{C} = \gamma_1 \cdot (\eta \cdot D), \quad (h \cdot \mathcal{C}) \cap F = \emptyset.$$

Applying h to the sunflower preserves disjointness of its $n + 1$ petals. At most n of those petals meet F , so one avoids F . Its core already avoids F , and for the corresponding index i we obtain

$$(h\gamma_i \cdot E) \cap F = \emptyset.$$

Finally,

$$|h\gamma_i| \leq |\gamma_1| + |\eta| + |\gamma_1^{-1}| + |\gamma_i| \leq 3R + 3(k-1)R = 3kR.$$

This proves (4.2). Taking $E = P$ and using the logarithmic bound on R proves the lemma. $\square$

Only the original points of P enter the growth estimates. The recursive step changes the forbidden set, but not its cardinality or the constants. This is why the same radius R works throughout the induction.

5. PACKING AND THE FINAL COMPARISON

Proof of Theorem 1.1. Suppose that Γ has rapid decay and trivial amenable radical but is not C^* -simple. Choose H and P from Lemma 2.2, and write $m = |P|$.

Fix $N \geq 2$. Begin with P itself. If j pairwise disjoint conjugates have been chosen, their union has jm points. Lemma 4.2 moves the original P off this union by a conjugator of length at most $C_P \log(jm + 1)$. Continuing in this way gives N pairwise disjoint conjugates

$$\gamma_1 P \gamma_1^{-1}, \dots, \gamma_N P \gamma_N^{-1}, \quad |\gamma_i| \leq D_P \log(N + 1),$$

where D_P depends only on P . Each step starts again from P , so the displacement constants do not change.

Put $L_P = \max_{p \in P} |p|$ and

$$r_N = \lceil L_P + 2D_P \log(N + 1) \rceil.$$

By (3.1), all the conjugates lie in B_{r_N} . Each meets H by (2.1), and their disjointness makes these intersection points distinct. Thus Lemma 2.3 and (1.1) give

$$N \leq |H \cap B_{r_N}| \leq \rho(r_N)^2 \leq C^2(1 + r_N)^{2d}. \quad (5.1)$$

Since $r_N = O_P(\log N)$, the right-hand side is bounded by a fixed power of $\log N$, a contradiction for large N . $\square$

Corollary 5.1. *A finitely generated group with sub-rapid decay and trivial amenable radical is C^* -simple.*

Proof. The proof above uses rapid decay only in the last inequality of (5.1). The construction still gives $N \leq \rho(r_N)^2$, where $r_N \rightarrow \infty$ and $r_N = O_P(\log N)$. Sub-rapid decay would therefore imply

$$1 \leq \frac{2 \log \rho(r_N)}{\log N} = 2 \frac{\log \rho(r_N)}{r_N} \frac{r_N}{\log N} \rightarrow 0,$$

which is impossible. $\square$

APPENDIX A. DISPLACEMENT AND PACKING CONSTANTS

We give the calculations behind the logarithmic estimates. They are useful for seeing exactly where the size of the fixed set enters; no optimization is intended.

Fix a finite nonempty $P \subseteq \Gamma \setminus \text{Rad}(\Gamma)$ and put $m = |P|$. Choose

$$c = \min_{p \in P} c_p \in (0, 1], \quad \delta = \min_{p \in P} \delta_p > 0$$

from Lemma 3.1. For a forbidden set of size $n \geq 1$, take

$$R(n) = 1 + \left\lfloor \frac{1}{\delta} \log \frac{m m! n^m}{c} \right\rfloor. \quad (A.1)$$

The expression inside the floor is nonnegative. The added one makes (4.1) strict, even when that expression is an integer. Also,

$$R(n) \leq 1 + \frac{\log(m m! / c)}{\delta} + \frac{m}{\delta} \log n.$$

The proof of Lemma 4.2 gives a conjugator of length at most $3mR(n)$. Since $\log n \leq \log(n + 1)$ and $\log(n + 1) \geq \log 2$, one may take

$$C_P = 3m \left\lceil \frac{m}{\delta} + \frac{1 + \delta^{-1} \log(m m! / c)}{\log 2} \right\rceil. \quad (A.2)$$

For $n = 0$, the identity has the required length zero.

To pack $N \geq 2$ conjugates, the largest forbidden set has $m(N - 1)$ points. Since $R(n)$ is nondecreasing, every conjugator can be chosen with length at most

$$T_N = 3mR(m(N - 1)).$$

Consequently,

$$T_N \leq K \log N + B, \tag{A.3}$$

where

$$K = \frac{3m^2}{\delta}, \quad B = 3m \left[1 + \frac{\log(m m! / c) + m \log m}{\delta} \right].$$

For example, $D_P = K + B / \log 3$ gives $T_N \leq D_P \log(N + 1)$ for all $N \geq 2$.

Put $L_P = \max_{p \in P} |p|$. The integer radius $L_P + 2T_N$ contains all the conjugate sets, and under the hypotheses of Theorem 1.1 the final comparison becomes

$$N \leq C^2 (1 + L_P + 2K \log N + 2B)^{2d}.$$

The right-hand side divided by N tends to zero. All constants here depend only on P and its rooted-growth constants, apart from the fixed rapid-decay constants C, d ; none depends on the positions of the forbidden points or on N .

USE OF LANGUAGE MODELS AND DEVELOPMENT OF THE PROOF

This disclosure concerns the alternative proof recorded in this supplement. I selected Question 1 of [AGKP] as a problem to investigate with a language model. The mathematical argument was generated by the model; I subsequently worked through it and verified it independently. Problem selection and independent verification were my mathematical contributions to this proof. I also requested changes to the exposition as I worked toward a version I could explain clearly: emphasizing the route from an amenable URS to a finite confinement set, explaining the role of short conjugators, and retaining the constants calculation.

The work was carried out in OpenAI's Codex environment. The mathematical worker agents were configured to use GPT-6 Astra with `xhigh` reasoning effort. Language-model assistance extended to proposing arguments, searching and reading literature, generating and revising LaTeX, and producing the Lean development described below. It was not limited to copyediting.

The research was organized around a coordinating agent and parallel worker agents. Different workers were given different tasks: pursuing a proof, looking for counterexamples, checking a proposed implication, or checking an external source. The coordinator collected their reports, compared the approaches, and retained the useful statements and unresolved objections. In this sense the work alternated between *fanout*, in which several approaches were explored, and *compression*, in which their conclusions were reduced to a smaller set of explicit proof obligations for the next stage. Summaries and handoff files also carried the state of the work across context limits. Those summaries were accompanied by saved proof versions, source records, and objections; the summaries themselves were not treated as proofs.

The route to the present argument was not immediate. The early work considered centralizers, stationary states, averaging estimates, and possible counterexamples. A useful reduction retained the whole amenable URS and its finite confinement set, rather than trying to settle the problem from the growth of one conjugacy class alone. The rooted-growth argument used the tuple mechanism already present in [AGKP], together with the observation that, for each element outside the amenable radical, some finite set of its conjugates generates a nonamenable subgroup. The worker who obtained the rooted-growth bound left the simultaneous finite-set step open. The coordinating model then proposed the sunflower displacement argument: many images give a proper core, induction moves the core, and a disjoint petal survives the forbidden set. Repeating the displacement gives the logarithmic packing used in the final contradiction.

An earlier mathematical compression argument required an additional hypothesis that was not derived from the full amenable-URS obstruction. That conditional argument is not part of this proof. It is separate from the compression of research reports and context described above. Later model-assisted revisions simplified the sunflower induction by pulling the core back into the original finite set. The present version makes the corresponding change to the forbidden set explicit and uses the bound $3kR$.

Candidate proofs were saved before being sent to other agents for criticism. The reviews checked the group and word-length hypotheses, the distinction between rapid and sub-rapid decay, the costs of moving the sunflower core, and the final counting argument. Their comments and revisions were retained. These were separate review tasks, but they used related models and shared mathematical sources. Agreement between them was not an independent mathematical certification. My own verification was a separate step.

A Lean formalization was also generated.¹ The retained records contain successful builds and axiom audits for a frozen development proving separate rapid-decay and sub-rapid-decay statements about the reduced group C^* -algebra. That development uses the original quantitative packing argument and a direct conjugation-averaging proof of simplicity; it does not formalize this revised TeX text line by line, and it does not use the later $3kR$ presentation. The build results apply to the saved formal sources and their definitions. They do not by themselves check the correspondence between every sentence of this exposition and the formal development.

After I communicated the existence of the generated proof and its Lean formalization to Joseph Gondek and Mehrdad Kalantar, we arranged a meeting at which I would present the argument. I began with the observation that, for an element outside the amenable radical, some finite set of its conjugates generates a nonamenable subgroup. At that point, the participants recognized that the observation was enough to obtain an answer by another argument. We stopped the presentation and agreed to prepare the joint Barnett–Gondek–Kalantar paper without using an LLM. That paper is the primary account of the result. This supplement records the earlier AI-generated alternative argument and its development. Gondek and Kalantar did not read or verify this alternative proof.

The present exposition, including this disclosure, was rewritten with language-model assistance. The account of my contributions and of the meeting is based on information I supplied. The account of the automated workflow is based on the retained research records. I take responsibility for the mathematical argument in this supplement and for the accuracy of this description.

REFERENCES

- [AGKP] N. Alam, J. Gondek, M. Kalantar, and R. Pham, *Growth conditions for topological freeness*, arXiv:2602.15009v2, 2026. <https://arxiv.org/abs/2602.15009v2>.
- [BHV] B. Bekka, P. de la Harpe, and A. Valette, *Kazhdan’s Property (T)*, New Mathematical Monographs, vol. 11, Cambridge University Press, Cambridge, 2008. Appendix G locators refer to the author manuscript of 23 February 2007, <https://perso.univ-rennes1.fr/bachir.bekka/KazhdanTotal.pdf>.
- [ER] P. Erdős and R. Rado, *Intersection theorems for systems of sets*, J. London Math. Soc. **35** (1960), 85–90. <https://doi.org/10.1112/jlms/s1-35.1.85>.
- [Ken] M. Kennedy, *An intrinsic characterization of C^* -simplicity*, Ann. Sci. Éc. Norm. Supér. (4) **53** (2020), no. 5, 1105–1119. Theorem numbering refers to arXiv:1509.01870v5, <https://arxiv.org/abs/1509.01870v5>.

¹The verification repository is <https://github.com/barnettcaleb1/q1-lean-verification>. The archived development uses Lean 4.33.1 and mathlib revision `0df444a360eaa60ab8c11dca51a86af692955474`. Its final declarations are `Q1.q1_rd` and `Q1.q1_srd`. These version identifiers refer to the saved formal development, not to a new Lean run for this rewrite.